Differential Equations

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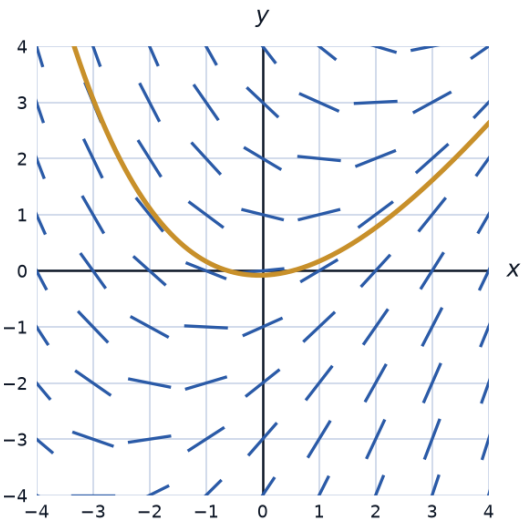
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

2 [Foundational] • Target: Forming differential equations from a description

The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.



3 [Foundational] • Target: Forming differential equations from a description

A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.

4 [Foundational] • Target: Forming differential equations from a description

A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.

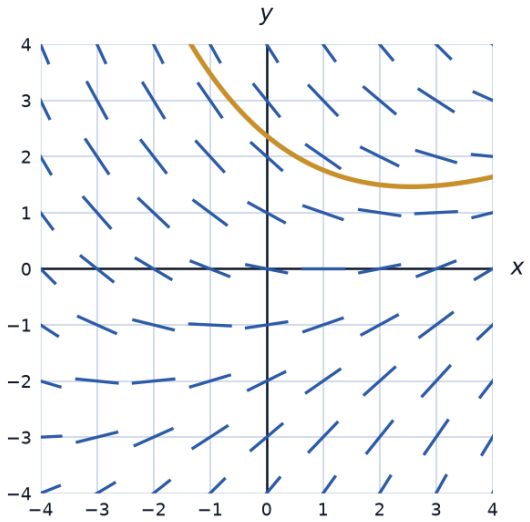
5 [Foundational] • Target: Solving modeling equations with initial conditions

A population P satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.

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6 [Foundational] • Target: Solving modeling equations with initial conditions

Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.



7 [Foundational] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.

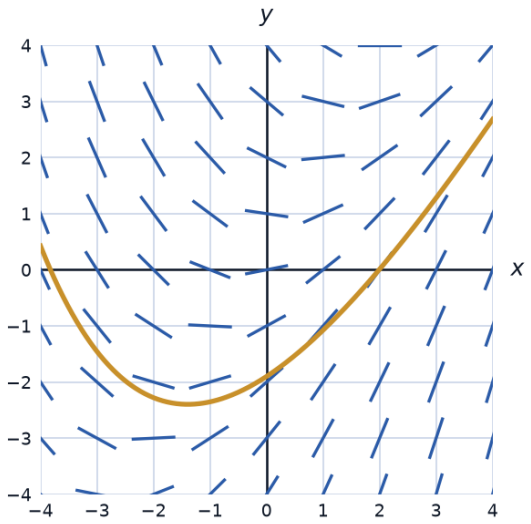
9 [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

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10 [Intermediate] • Target: General and particular solutions

Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.



11 [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

12 [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ and passes through $(1, 4)$. Find the particular solution.

13 [Intermediate] • Target: Applications: Newton's law of cooling

A cup of tea at $85\,^\circ$C is placed in a room at $22\,^\circ$C. Using $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 8 minutes the tea has cooled to $65\,^\circ$C, find k, to three decimal places.

14 [Intermediate] • Target: Applications: Newton's law of cooling

Using the value of k found above, find the temperature of the tea after 20 minutes, to the nearest degree.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

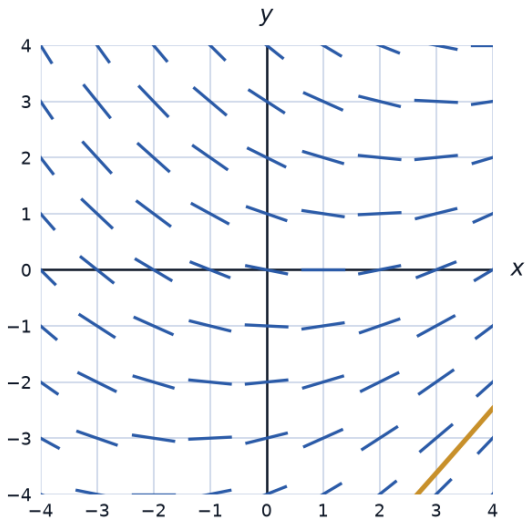
15 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 12.01 Poisson Distribution

A metal rod at $200\,^\circ$C cools in air at $25\,^\circ$C, with cooling constant $k = 0.03$ per minute. Find its temperature after 15 minutes, to the nearest degree.

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16 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 13.01 Continuous Random Variables

This question has three parts. A tank contains 300 litres of brine with 10 kg of dissolved salt. Fresh water flows in at 6 litres/minute, and the well-mixed solution flows out at the same rate, keeping the volume constant. Let $S(t)$ be the mass of salt (kg) at time t minutes. (a) Explain why $\frac{dS}{dt} = -\frac{S}{50}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 10$. (c) Find how long it takes for the salt content to drop to 2 kg, to the nearest minute.



17 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 14.01 Sampling & Estimation

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

18 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 15.01 Hypothesis Testing

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

19 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 2.01 Exponentials & Logarithms

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

20 [Challenging] • Target: Forming differential equations from a description • Cross-Lesson: 3.01 Advanced Trigonometry

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Solving by separation of variables

First-order linear ODE solution structure: general solution = homogeneous solution + ____

a. particular solution

b. another homogeneous solution

c. the initial condition

d. zero

22 [Foundational] • Target: Forming differential equations from a description

A radioactive substance has half-life 2 years and initial mass 16g. Find the mass remaining after 4 years.

a. 4

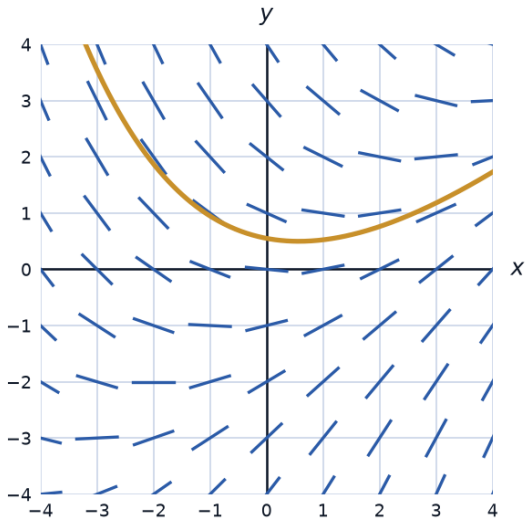
b. 16

c. 8

d. 5

23 [Foundational] • Target: Solving modeling equations with initial conditions

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):



a. $y = \frac{1}{2}x^3 + Cx$

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

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24 [Foundational] • Target: First-order linear differential equations using an integrating factor

Coffee at 70°C is placed in a 22°C room and cools to 45°C in 5 min (Newton's law of cooling). How long (min) until it reaches 35°C?

a. 7

b. $\frac{10\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

c. $\frac{5\ln(2)}{\ln\left(\frac{14}{9}\right)}$

d. $\frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

25 [Foundational] • Target: Further integrating factor examples

Newton's law of cooling: $\frac{dT}{dt} = k(T - T_s)$, $k < 0$. Steady state:

a. $T = 0$

b. $T = kT_s$

c. $T = T_s$

d. $T = T_0$

26 [Foundational] • Target: Verifying solutions to differential equations

A first-order linear ODE has the form:

a. $y' + P(x)y = Q(x)$

b. $y' = P(x)y^2$

c. $y'y = Q(x)$

d. $y'' = P(x)$

27 [Foundational] • Target: General and particular solutions

For the linear ODE $y' + 2y = 4$: integrating factor $\mu =$

a. e^2

b. 2

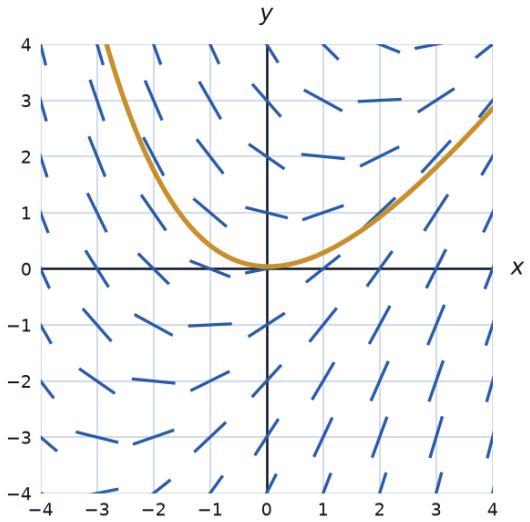
c. e^{-2x}

d. e^{2x}

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28 [Foundational] • Target: Applications: Newton's law of cooling

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:



a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a mixing tank

Which differential equation correctly models the statement: "The rate of increase of a population P is proportional to P itself."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dm}{dt} = -km$

d. $\frac{dN}{dt} = kN(S - N)$

30 [Intermediate] • Target: Solving by separation of variables

Find the general solution of $\frac{dy}{dx} = 2x^2$.

a. $y = \frac{2x^3}{3} + C$

b. $y = \frac{2x^3}{3}$

c. $y = 6x^3 + C$

d. $y = 2x^2 + C$

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31 [Intermediate] • Target: Forming differential equations from a description

A quantity satisfies $\frac{dy}{dt} = -y$ with $y(0) = 9$. Find y when $t = \ln(3)/1$.

a. 4

b. 27

c. 3

d. 9

32 [Intermediate] • Target: Solving modeling equations with initial conditions

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

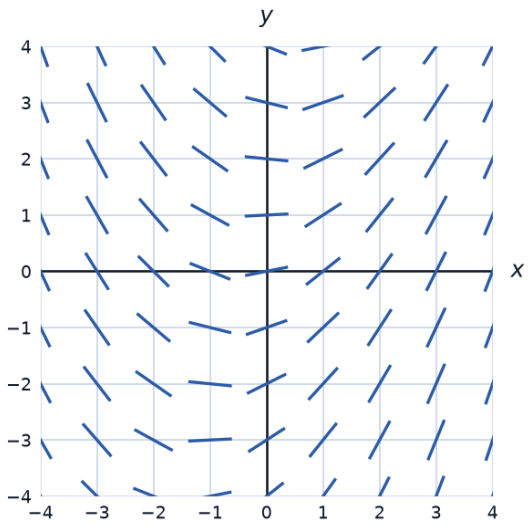
b. 1

c. $1 + e^2$

d. 2

33 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.



a. $e^{-1} + 1$

b. $2e^{-1} + 1$

c. $e^{-1} - 1$

d. $2 - e^{-1}$

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34 [Intermediate] • Target: Further integrating factor examples

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$

35 [Intermediate] • Target: Verifying solutions to differential equations

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$

36 [Intermediate] • Target: General and particular solutions

A 200-L tank holds brine with 20 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 20 min?

a. $100 - \frac{80}{e}$

b. $100 - \frac{100}{e}$

c. 120

d. $\frac{20}{e} + 100$

37 [Intermediate] • Target: Applications: Newton's law of cooling

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$

d. $-\cos x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 15.01 Hypothesis Testing

Integrating factor for $y' + P(x)y = Q(x)$:

a. $e^{P(x)}$

b. $e^{\int P\,dx}$

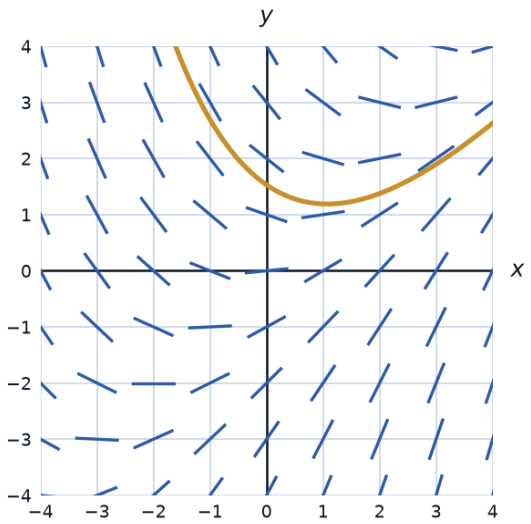
c. $P(x)$

d. $1/P(x)$

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39 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 2.01 Exponentials & Logarithms

Initial value problem: $y' = y$, $y(0) = 2$. Solution:



a. $y = e^{2x}$

b. $y = 2e^{x}$

c. $y = 2x + 2$

d. $y = xe^{x}$

40 [Challenging] • Target: Forming differential equations from a description • Cross-Lesson: 3.01 Advanced Trigonometry

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

41 [Challenging] • Target: Solving modeling equations with initial conditions • Cross-Lesson: 4.01 Differentiation

Fundamental theorem for linear ODEs: the general solution of $Ly = g$ is:

a. y_p only

b. $y_h + y_p$

c. y_h only

d. any solution of $Ly = 0$

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42 [Challenging] • Target: First-order linear differential equations using an integrating factor • Cross-Lesson: 5.01 Integration

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

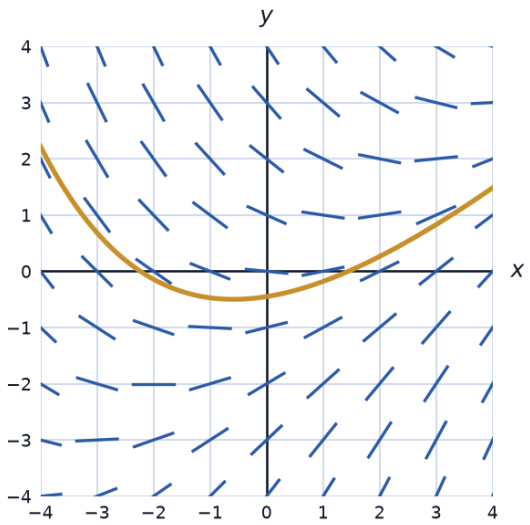
b. 45

c. 50

d. 54

43 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 6.01 Sequences & Series

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.



a. 32

b. -1

c. $\frac{32}{3}$

d. 16

44 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 7.01 Vectors

The continuous random variable X has probability density function given by

$f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise},$ where k is a constant. Find the value of k.

a. $\frac{3}{16}$

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{16}{3}$

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45 [Challenging] • Target: General and particular solutions • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128

b. 96

c. 74

d. 64

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